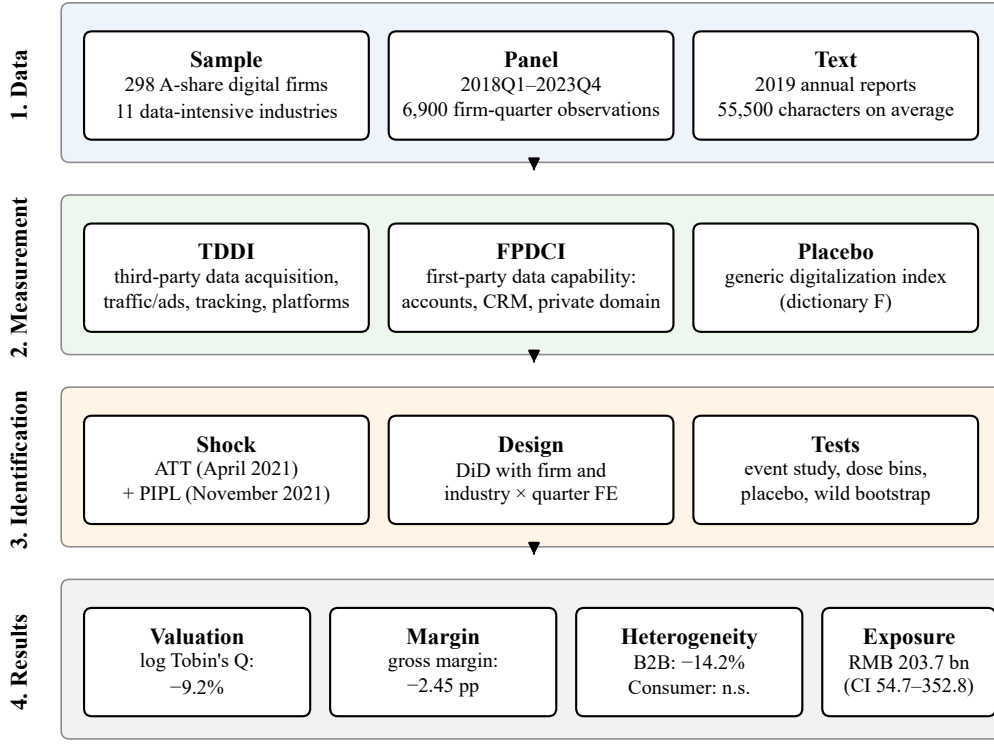

1. Data
Sample
298 A-share digital firms
11 data-intensive industries
Panel
2018Q1–2023Q4
6,900 firm-quarter observations
Text
2019 annual reports
55,500 characters on average
2. Measurement
TDDI
third-party data acquisition,
traffic/ads, tracking, platforms
FPDCI
first-party data capability:
accounts, CRM, private domain
Placebo
generic digitalization index
(dictionary F)
3. Identification
Shock
ATT (April 2021)
+ PIPL (November 2021)
Design
DiD with firm and
industry × quarter FE
Tests
event study, dose bins,
placebo, wild bootstrap
4. Results
Valuation
log Tobin's Q:
−9.2%
Margin
gross margin:
−2.45 pp
Heterogeneity
B2B: −14.2%
Consumer: n.s.
Exposure
RMB 203.7 bn
(CI 54.7–352.8)